

GENDER BIAS IN HEALTHCARE: MISSED APPOINTMENT PREDICTION

Artificial Intelligence Ethics and Applications

CIS4057-N-FJ1-2025

JENNIFER ROBERTS

E5147249

2443 words

Introduction

Bias includes anything that causes people to be treated unfairly. Discrimination resulting from bias related to certain characteristics – including sex (referred to as gender in this paper) – is prohibited by the *Equality Act 2010*.

It is well-established that gender bias exists in healthcare at all levels, starting with medical research (Holdcroft, 2007; Yoon *et al.*, 2014; Cirillo *et al.*, 2020). Artificial intelligence can reflect and amplify this bias, and every step of development can introduce different types (Suresh & Guttag, 2021; Nazer *et al.*, 2023; Hasanzadeh *et al.*, 2025).

If not addressed, this bias typically results in women having poorer healthcare outcomes than men due to misdiagnosis, late diagnosis, and/or ineffective treatment. For example, Cao *et al.* (2022) showed that a model developed by the US Department for Veterans Affairs and Deepmind to detect acute kidney injury performed significantly worse on women than men. Meanwhile, Rickman (2025) found that Google's Gemma LLM tended to downplay women's mental and physical health needs when used to produce long-term care notes.

Predicting missed appointments may seem comparatively unimportant. However, even in this area biased models can compound existing inequalities (Samorani & Goler Blount, 2020; Parsons, Bryce & Atherton, 2021; Samorani *et al.*, 2021; Roberts & Salib, 2024). Many providers double-book appointments to compensate for predicted no-shows. This reduces wasted time for clinicians, but leads to increased waiting times and/or rushed appointments if people do attend. This may mean they leave without being seen, or avoid making further appointments, and lead to worse health outcomes.

The limited previous work in this area has focused on racial bias; this work will consider gender.

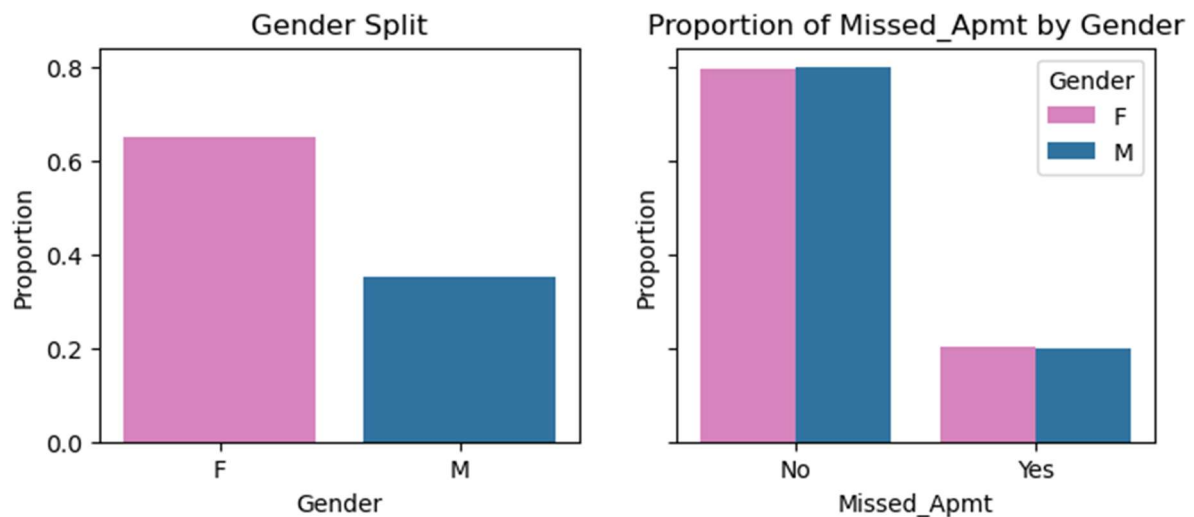
Model Development

The dataset of medical appointments used contains fourteen features and 110,527 instances. Each instance is an individual appointment, although patients may have more than one listed. The features include patient demographics and health conditions, the appointment location, booking and appointment dates, and whether they were sent a text message reminder. Inspection revealed no null values or duplicate rows. Several features were renamed for clarity.

Columns were checked for outliers; three rows were removed due to incorrect age values and 221 rows from seven neighbourhoods with less than one hundred appointments each (listed in the model card). This was to avoid chance patterns in limited data reducing the usefulness of the Neighbourhood variable. However, it does limit the utility of the overall model.

The target variable is whether they missed the appointment or not. It is unbalanced, with about 20% in the minority class (missed appointment). The overall gender distribution of patients is 64% female and 36% male, and of appointments is 65% female and 35% male. The target variable proportions are approximately the same for both.

Figure 1: Gender split, and proportion of missed appointments by gender.



Exploratory Data Analysis (EDA) showed gender differences in several variables, particularly age group (Figure 2) and neighbourhood (Figure 3), although none could be considered proxies for gender.

Figure 2: Gender balance of appointments by age group.

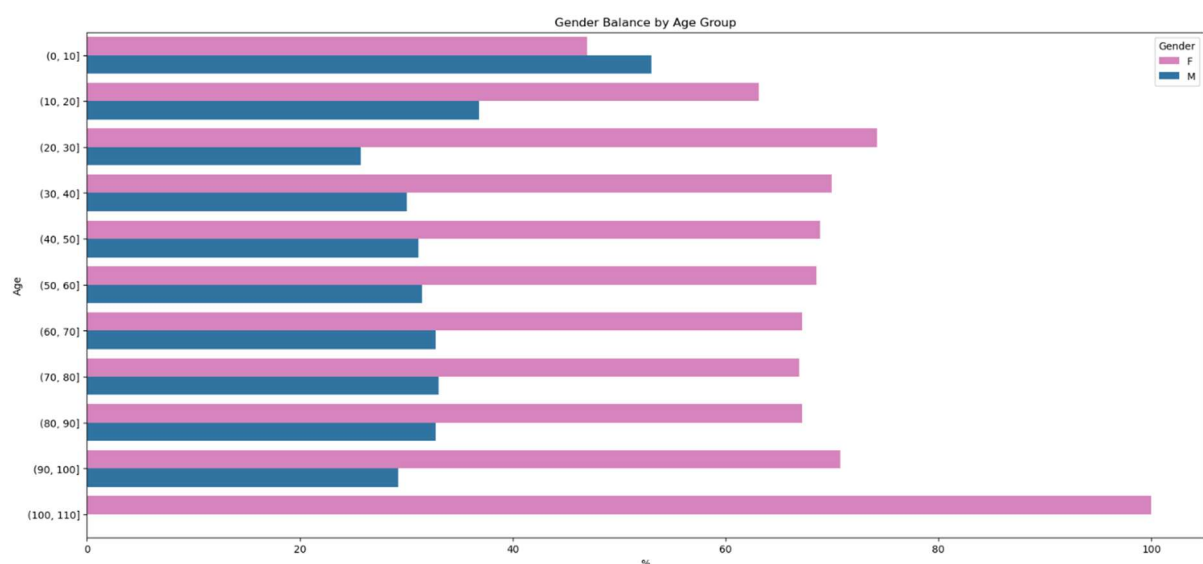
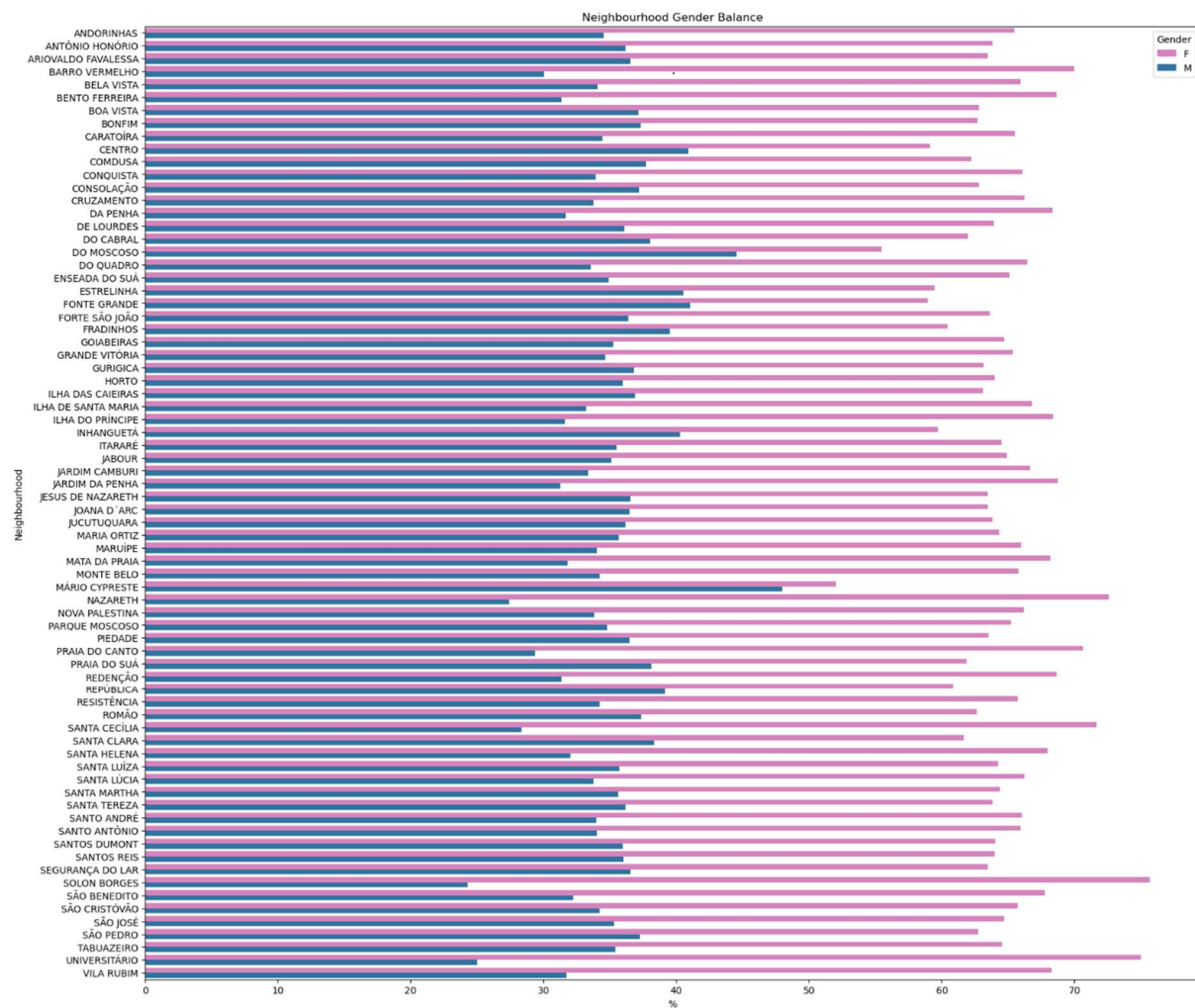


Figure 3: Gender balance of appointments by neighbourhood.



There was also some variation in the percentage of missed appointments by age group (Figure 4) and neighbourhood (Figure 5).

Figure 4: Proportion of missed appointments by age group.

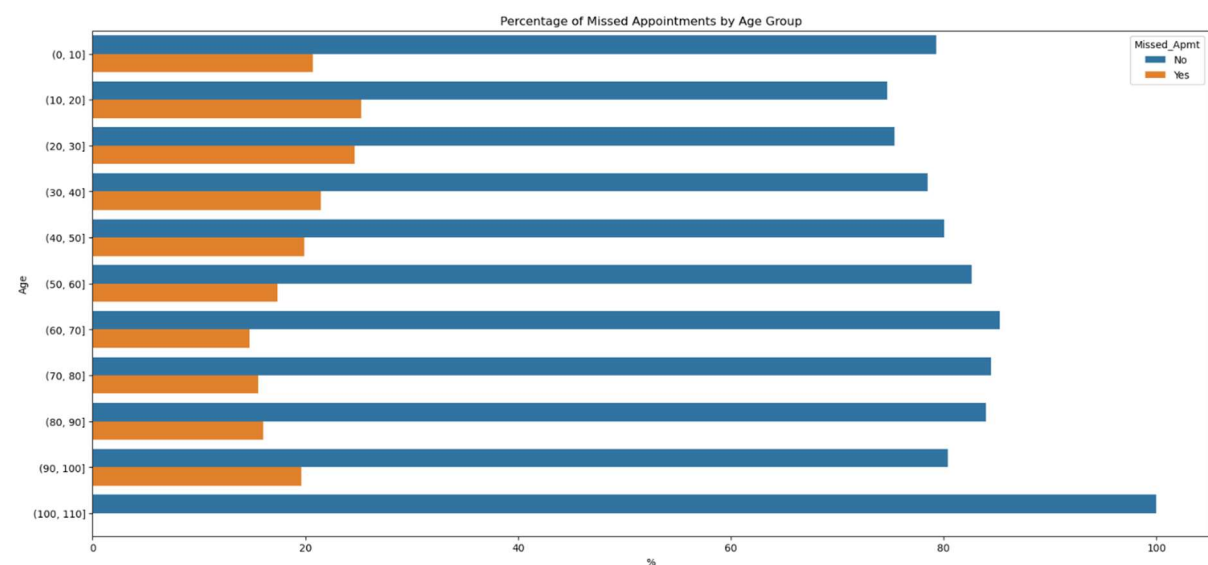
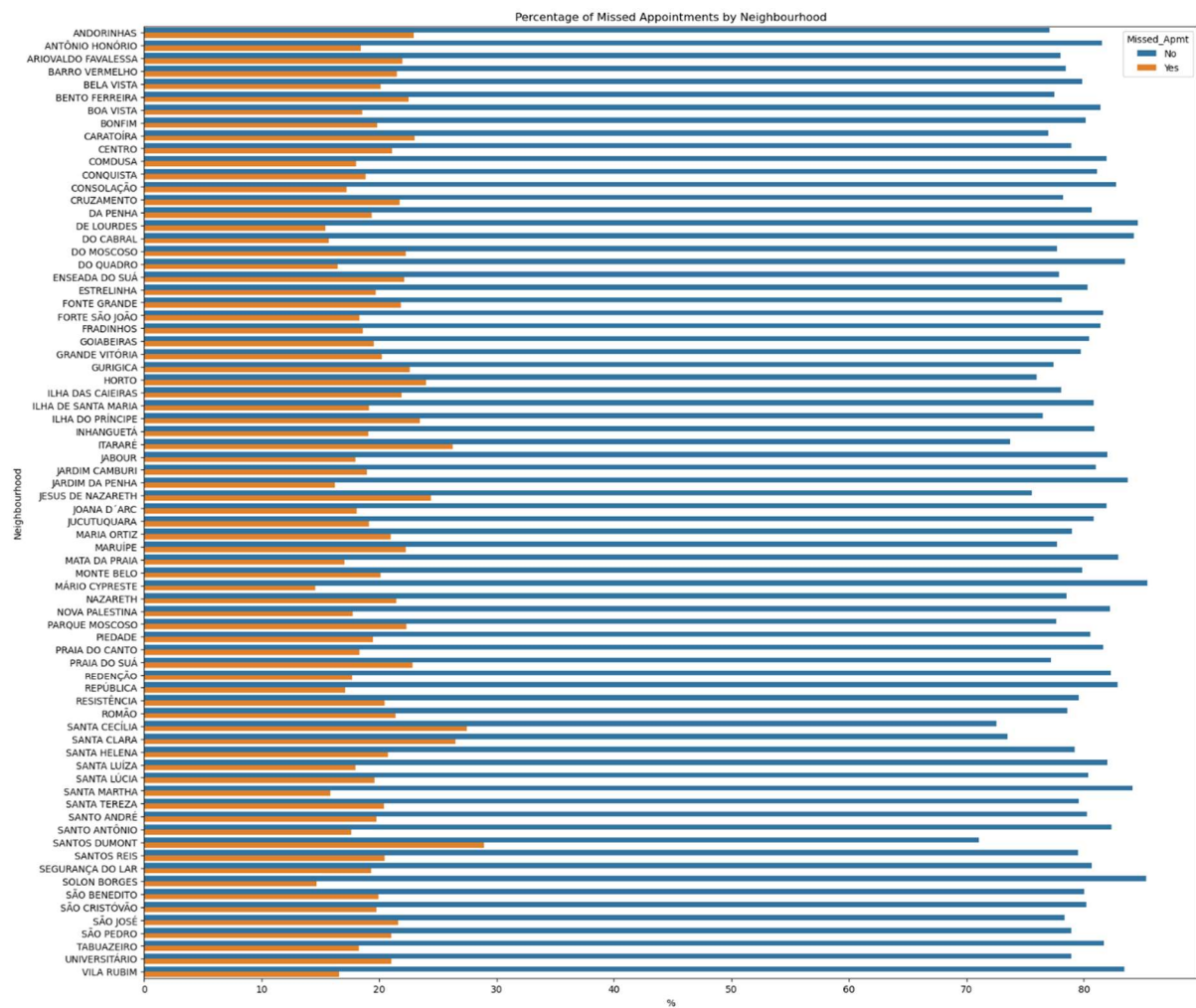
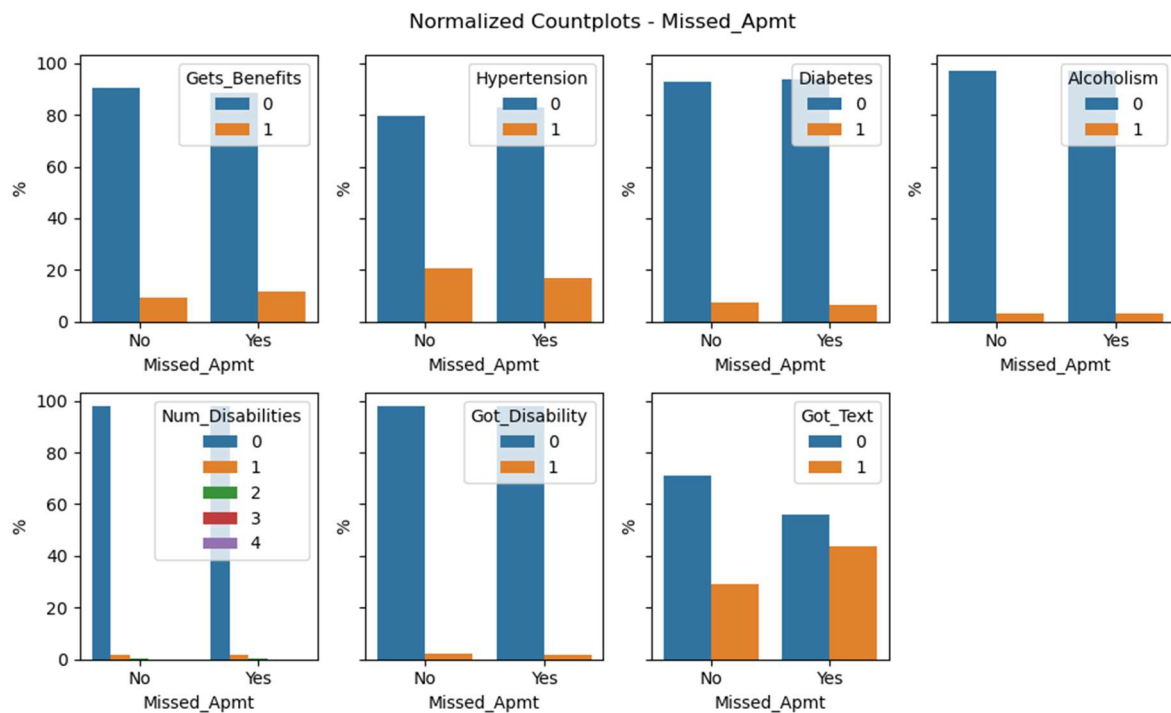


Figure 5: Proportion of missed appointments by neighbourhood.



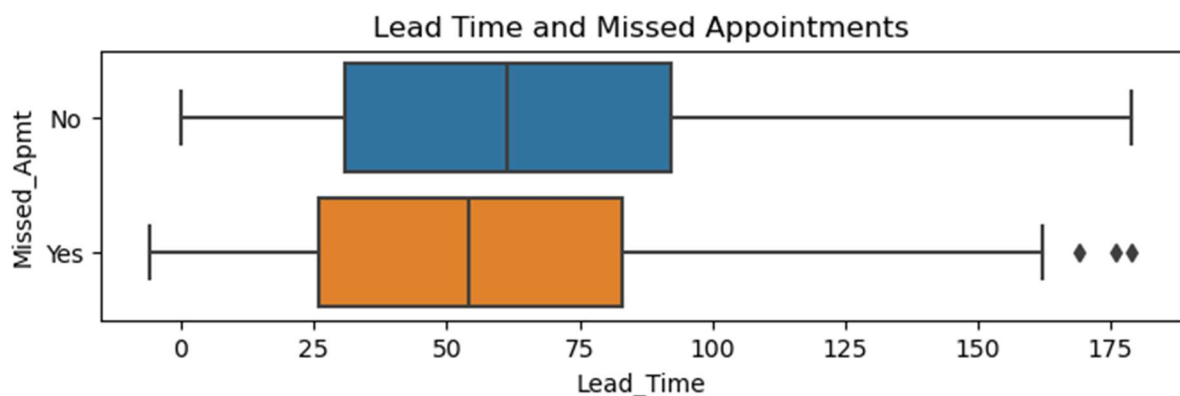
Of the other variables, whether the patient was sent a text appears to be the most important.

Figure 6: Proportion of missed appointments by various variables.



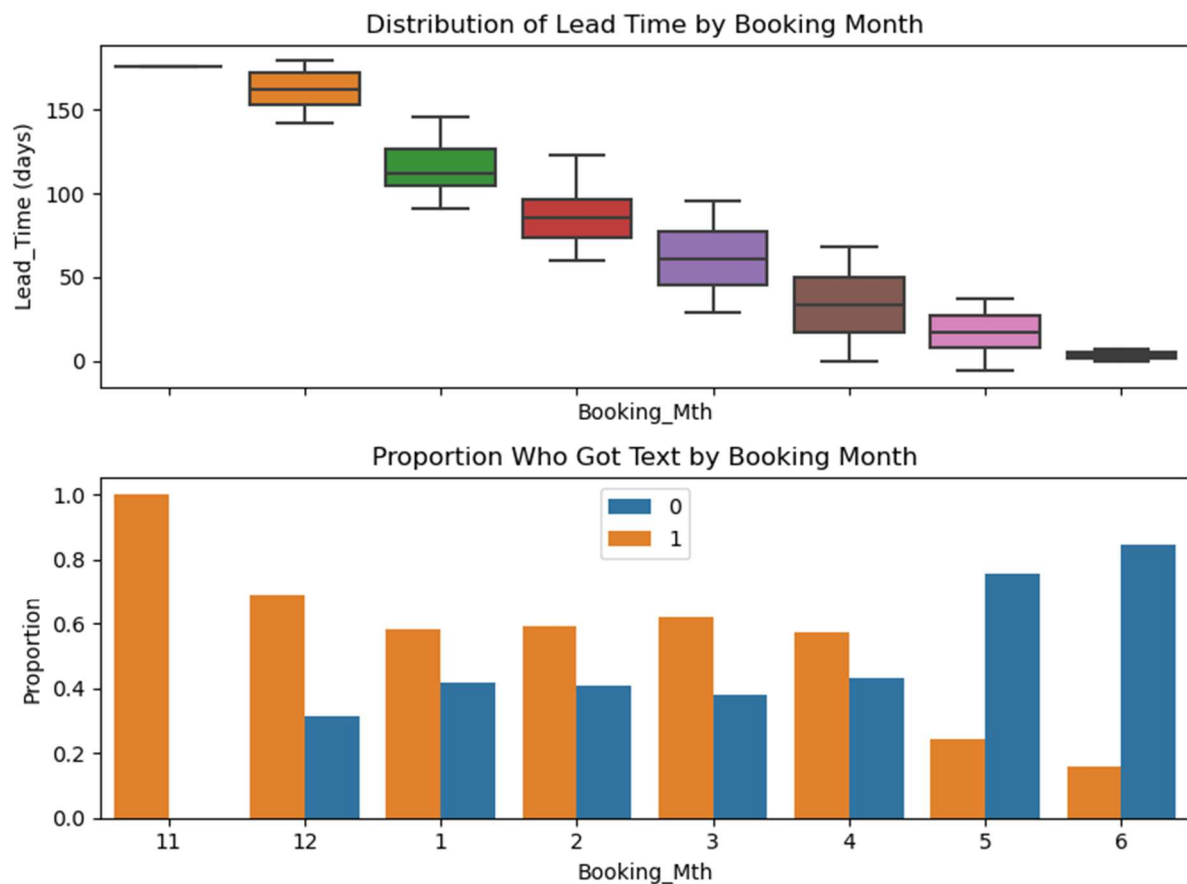
Appointment and booking dates were split into month, day, and day-of-the-week. Appointment hour was discarded, as all were zero. It was found that booking dates spanned seven months, but appointments only five weeks. Consequently, the appointment date was only used to calculate “lead time” (the time between booking and appointment – Parsons, Bryce & Atherton, 2021), as there was not enough data to find reliable patterns.

Figure 7: Distribution of lead time for missed and not missed appointments.



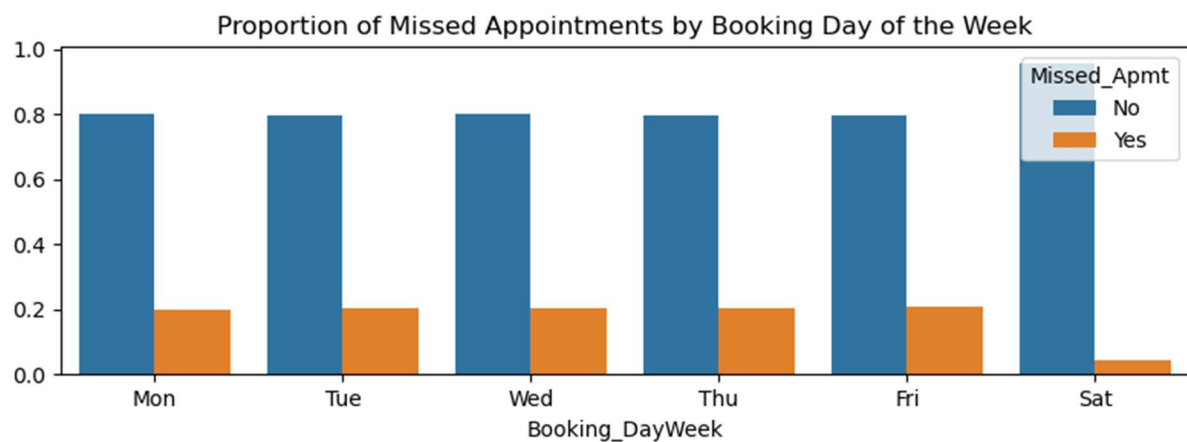
Booking month was subsequently dropped, because lead time and the likelihood of getting a text varied significantly (Figure 8).

Figure 8: Variation of lead time and got text with booking month.



Appointments booked on Saturdays appeared to be missed at a lower rate than weekdays (Figure 9). However, further investigation found that there were only twenty-four Saturday bookings, across thirteen of the seventy-four neighbourhoods, so this is unlikely to be a reliable pattern.

Figure 9: Proportion of missed appointments by booking day of the week.



As none of the variables are continuous and most are categorical, chi-square and mutual information scores with Missed_Apmt (Figure 10) were computed for feature selection. The chi-square test found that the likelihood of missing an appointment is

independent of gender, alcoholism, number of disabilities and which day of the week the booking was made ($p = 0.05$). This suggests that these features should not be used in the model. Mutual information broadly agreed on the least important features. It did rate booking day-of-the-week considerably higher than chi-square, but as discussed above this variation seems to be unreliable. Meanwhile, MI ranked hypertension much lower than Chi-square did. Figure 6 showed that it only had a very small effect on missing appointments. Neither ranked neighbourhood particularly high, although Figure 5 did suggest some difference.

Both this and mutual information score identified booking day and whether the patient got a text as important (chi-square cannot be used for lead time and age, as they are not categorical). However, there is a risk that the Saturday variation may also be exaggerating the relevance of booking day.

Figure 10: Features selected by Chi-Square and Mutual Information scores.

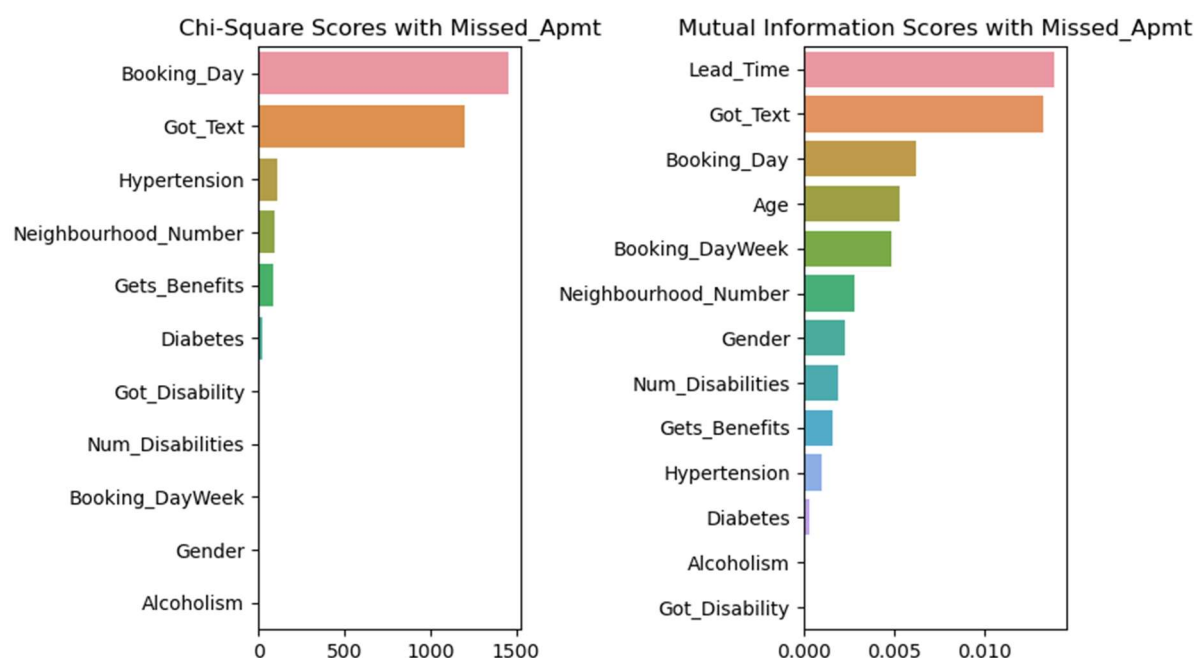


Figure 1 and the Chi-square test suggest that, in this dataset, gender is not related to likelihood of missing an appointment (although mutual information is less conclusive). However, it is required to allow application of fairness criteria.

Text variables (Neighbourhood, Gender and Missed_Apmt) were label-encoded and non-ordinal variables were given type category. The dataset was split 70:30 into training and testing sets, using stratification to preserve the target variable proportions. The training set was then balanced using random oversampling of the missed appointments to preserve as much data as possible. The training and test sets (minus the target variable) were scaled using a Robust Scaler (as the most resistant to the outliers seen in lead time) scaled on the training set (the same was used for both to maintain consistency). Random state 123 was used throughout for reproducibility.

Based on the above, the features selected for the model were: Lead_Time, Got_Text, Booking_Day, Age, Neighbourhood_Number and Gender, with Missed_Apmt as the target variable.

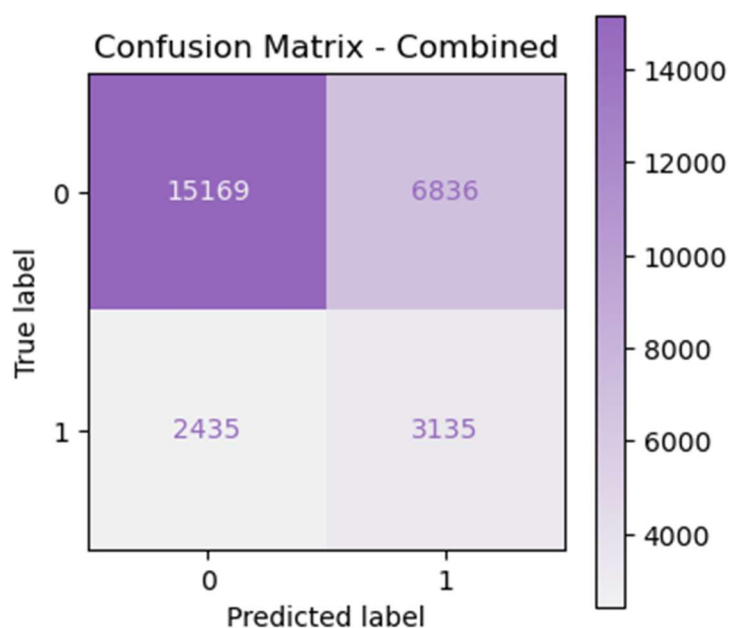
The model used was Logistic Regression, which is both robust and fast and provides a good baseline model. Optimal C value was found by a grid search with 5-fold cross validation and Precision as the scoring factor. Precision was chosen because it considers the false positive rate, which the model aims to minimise.

The final model metrics are displayed in Table 1 (as the model is not going into production, cross-validation was not performed on these scores).

Table 1: Model metrics (overall) on testing data.

	Accuracy	Precision	Recall
Result	0.66	0.31	0.56

Figure 11: Confusion matrix (overall) on testing data.



Despite best efforts at feature engineering and selection, this model is not strong (ironically, it is particularly weak on Precision) and should not be deployed in its current state. Variables that might produce a better model include how far they need to travel to the appointment, what forms of transport they have access to and whether they have missed an appointment before (Parsons, Bryce & Atherton, 2021).

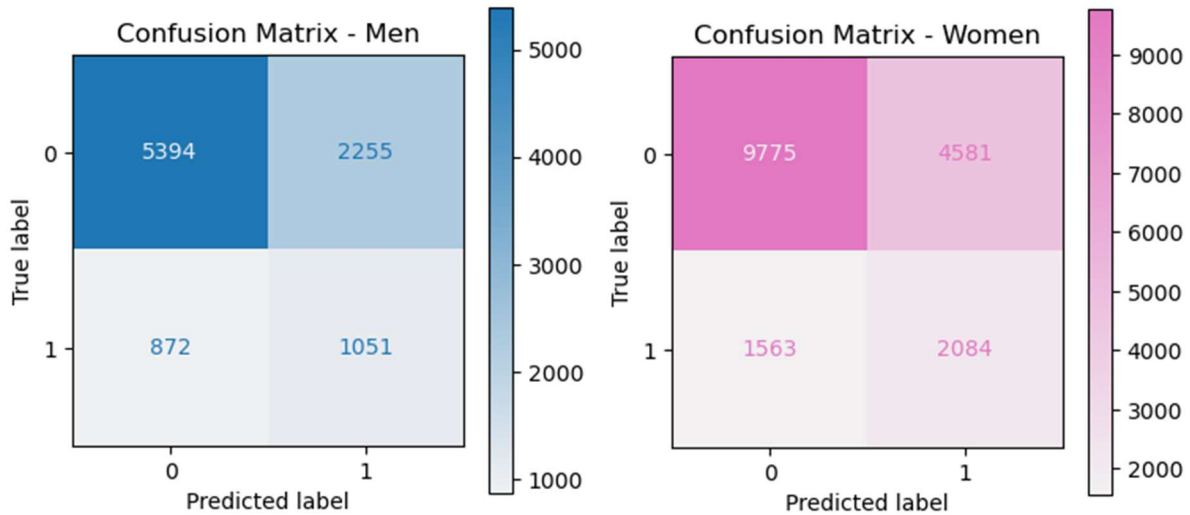
Application of Fairness Criteria

The model predictions were then divided by gender to allow comparison.

Table 2: Model metrics by gender on test data.

	Accuracy	Precision	Recall
Men	0.67	0.32	0.55
Women	0.66	0.31	0.57

Figure 12: Confusion matrices by gender.



Based on the model metrics (Table 2) and the confusion matrices (Figure 12), there is no obvious gender bias in the model. However, that does not guarantee that it is fair. Bias was checked using Equal Accuracy, Demographic Parity and Equal Opportunity tests.

Equal Accuracy requires that the model be equally accurate for the protected and privileged groups (female and male here). However, Accuracy only considers the number of errors and not the nature of them. Accuracy is also dependent on the “ground truth” (the target variable) and so will reflect any societal bias in the original data (Barocas *et al.*, 2017, cited in Braun, 2024). It is not commonly used to assess fairness and so there is no established threshold to satisfy it. The same threshold as for Demographic Parity has been used here.

Demographic Parity requires that the rate of positive prediction be the same for both groups (i.e. that men and women are equally likely to be predicted as missing their appointment). However, it still does not consider whether those positives are true or false; one group could still have a much higher proportion of false positives than the other (so it is not selecting those most likely to miss their appointment) (Gajane & Pechenizkiy, 2018). The most used threshold is from the Disparate Impact Liability guidelines (USA), which say the positive prediction rate for the protected group should be at least 80% of the rate for the privileged group.

In this dataset, gender does not affect the chance of missing an appointment, so Demographic Parity is both achievable and desirable. However, if women were more

likely to miss their appointment – e.g. in a country where they need a male escort to travel – achieving Demographic Parity would require “reverse discrimination”: either underpredicting for women or overpredicting for men (Braun, 2024). This would reduce the model’s overall accuracy.

Equal Opportunity instead considers the true positive rate (TPR). In this model, that means that an equal proportion of women and men who miss their appointments are correctly identified. This is helpful, but it still does not guarantee an equal false positive rate. The threshold used here is the Equal Opportunity Difference (EOD), subtracting the TPR/Recall of the protected group from that of the privileged group.

Like Accuracy, EO is dependent on the ground truth data. As Mehrabi et al. (2021) point out, fairness criteria tend to focus on equality rather than equity, ignoring the often-complex history beneath said ground truth. According to Hardt, Price & Srebro (2016), forcing a model to satisfy Equal Opportunity should not affect accuracy as much as Demographic Parity does.

The results of these tests on this model can be seen in Table 3.

Table 3: Fairness test results.

	Men	Women	Threshold	Actual Value	Passed?
Equal Accuracy	0.67	0.66	N/A	0.01	Yes
Demographic Parity	0.35	0.37	Ratio ≥ 0.80	1.07	Yes
Equal Opportunity	0.55	0.57	Difference < 0.05	-0.02	Yes

There is no agreed definition of fairness in machine learning (Mehrabi, 2021) and no test for which passing guarantees that the model is fair. Some proposed fairness criteria are even mutually exclusive (Kleinberg, Mullainathan & Raghavan, 2016, cited in Mehrabi *et al.*, 2021). The selection of appropriate tests depends on domain knowledge (Hardt, Price & Srebro, 2016), to determine whether false positives or false negatives have more serious consequences.

In this use case, correct predictions are mostly neutral. A high proportion will minimise wasted time and have a universal positive effect on clinic and patients, but should not impact the individual patient at all.

False negatives are neutral for the individual patient (unless the clinic issues penalties for non-attendance). They would create a gap in the schedule which would reduce the number of patients the clinic sees that day, which could affect evaluation metrics and/or state funding. It could also increase the wait to book appointments. If the clinic is running on time, it also means wasted time for the individual clinician. On the other hand, if the clinic is running late this gives them a chance to catch up, which could provide a better experience for the other patients waiting. It is false positives that could

cause disparate impact on protected groups. Given this, Equalized Odds (Hardt, Price & Srebro, 2016) might be a more suitable fairness test (in fact, it passes this too).

A model can also be “fair” for large groups but unfair to sub-groups (Mehrabi *et al.*, 2021). Given the gender variation by age group (Figure 2), it could be that the model fails fairness tests for women in a particular age group although it passes them for women overall. All the above tests measure group-based fairness, so the model may still be unfair to individuals.

Finally, a model can start off fair, but become unfair over time due to model drift (Nazer *et al.*, 2023).

Findings

The model passed all three fairness tests at the thresholds set, so no bias was detected, but that does not mean it is fair. First, there are many proposed tests of fairness and this paper has only used three. Second, only gender fairness is considered here; the model could be biased with respect to e.g. age or race.

Assume there is bias in the model. It is false positives that cause a problem for the individual. In this case, their appointment will be double-booked. Their appointment may feel rushed, reducing their quality of care. They may have a longer wait, and may even have to leave without being seen due to other commitments (the same would go for other waiting patients). All of these may discourage them from seeing the doctor again (Roberts & Salib, 2024). As a result, they may suffer ill-health unnecessarily, which will have a negative impact on all aspects of their life. Serious conditions may not be diagnosed until a late stage, requiring longer and/or more aggressive treatment and decreasing their chance of survival or recovery. If the model predicted significantly more false positives for women than men, this bias would certainly cause an impact.

Roberts & Salib (2024) suggest a policy of seeing the affected patient first at their booked time could mitigate this. However, there is also the question of the patient who is booked second into the slot; is it fair to expect them to wait? And are they just next on the list, or is the model’s prediction being considered for them as well? If a clinic pairs predicted no-shows, assuming that one of the two will keep their appointment, any bias in false positives will double.

In summary, ascertaining the existence of bias in a model is extremely difficult and models can impact social groups in complicated and wide-reaching ways. Consequently, the ethical implications of using similar models in such a high-risk setting as healthcare should be carefully considered before implementation.

References

- Braun, C. (2024) 'Fairness in Machine Learning.' *dida blog*, 3 Jan. Available at: <https://dida.do/blog/fairness-in-ml>. (Accessed: 27 April 2026).
- Cao, J., Zhang, X., Shahinian, V., Yin, H., Steffick, D., Saran, R., Crowley, S., Mathis, M., Nadkarni, G.N., Heung, M., Singh, K. (2022) 'Generalizability of an acute kidney injury prediction model across health systems.' *Nature Machine Intelligence* 4(12), pp. 1121-1129. doi: 10.1038/s42256-022-00563-8.
- Cirillo, D., Catuara-Solarz, S., Morey, C., Guney, E., Subirats, L., Mellino, S., Gigante, A., Valencia, A., Rementeria, M.J., Chadha, A.S. and Mavridis, N. (2020) 'Sex and gender differences and biases in artificial intelligence for biomedicine and healthcare', *NPJ Digital Medicine*, 3, doi: 10.1038/s41746-020-0288-5
- Cross, J.L., Choma, M.A. and Onofrey, J.A. (2024) 'Bias in medical AI: Implications for clinical decision-making.' *PLOS Digital Health*, doi: 10.1371/journal.pdig.0000651
- Equality Act 2010* c.15. Available at: <https://www.legislation.gov.uk/ukpga/2010/15/contents> (Accessed: 25 April 2026).
- Gajane, P. and Pechenizkiy, M. (2018) 'On Formalizing Fairness in Prediction with Machine Learning.' *arXiv preprint*. Available at: arXiv:1710.03184
- Hardt, M., Price, E. and Srebro, N. (2016) 'Equality of Opportunity in Supervised Learning.' arXiv. Available at: <https://arxiv.org/pdf/1610.02413>
- Hasanzadeh, F., Josephson, C.B., Waters, G., Adedinsewo, D., Azizi, Z. and White, J.A. (2025) 'Bias recognition and mitigation strategies in artificial intelligence healthcare applications', *NPJ Digital Medicine*, 8(1), doi: 10.1038/s41746-025-01503-7
- Holdcroft, A. (2007) 'Gender bias in research: how does it affect evidence based medicine?' *Journal of the Royal Society of Medicine*, 100(1), doi: 10.1177/014107680710000102
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A. (2021) 'A Survey on Bias and Fairness in Machine Learning.' *ACM Computing Surveys* 54(6), doi: 10.1145/3457607
- Nazer, L.H., Zatarah, R., Waldrip, S., Ke, J.X.C., Moukheiber, M., Khanna, A.K., Hicklen, R.S., Moukheiber, L., Moukheiber, D., Ma, H. and Mathur, P. (2023) 'Bias in artificial intelligence algorithms and recommendations for mitigation.' *PLOS Digital Health* 2(6), doi: 10.1371/journal.pdig.0000278
- Parsons, J., Bryce, C. and Atherton, H. (2021) 'Which patients miss appointments with general practice and the reasons why: a systematic review.' *British Journal of General Practice* 2021, doi: 10.3399/BJGP.2020.1017

Rickman, S. (2025) 'Evaluating gender bias in large language models in long-term care.' *BMC Medical Informatics and Decision Making* 25, doi: 10.1186/s12911-025-03118-0

Roberts, J. L. and Salib, P. (2024) 'Algorithmic discrimination and health equity', in B. Solaiman and I.G. Cohen (eds) *Research Handbook on Health, AI and the Law*. Cheltenham, UK: Edward Elgar Publishing Ltd, pp. 92–110.

Samorani, M. and Goler Blount, L. (2020) 'Machine Learning and Medical Appointment Scheduling: Creating and Perpetuating Inequalities in Access to Health Care.' *American Journal of Public Health*, 110(4), pp. 440-441. doi: 10.2105/AJPH.2020.305570

Samorani, M., Harris, S.L., Blount, L.G., Lu, H. and Santoro, M.A. (2021) 'Overbooked and overlooked: machine learning and racial bias in medical appointment scheduling.' *Manufacturing & Service Operations Management*. doi: 10.2139/ssrn.3467047

Suresh, H. and Gutttag, J.V. (2021) 'A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle.' *arXiv*. doi: 10.48550/arXiv.1901.10002

Yoon, D.Y., Mansukhani, N.A., Stubbs, V.C., Helenowski, I.B., Woodruff, T.K. and Kibbe, M.R. (2014) 'Sex bias exists in basic science and translational surgical research.' *Surgery*, 156(3), pp. 508-516. doi: 10.1016/j.surg.2014.07.001

Appendix

Model Card

MODEL DETAILS	
Developed by: Jennifer Roberts Model date: 29 April 2026 Model version: 1.0 Model type: Logistic regression classifier using C = 0.01 and oversampling. No fairness constraints applied. Features:	
FEATURE	DESCRIPTION
Age	Age (years)
Booking_Day	The day of the month (number) when the booking was made.
Gender	Male/Female
Got_Text	Whether one or more text messages were sent to the patient about the appointment.
Lead_Time	Time between booking date and appointment date (days).
Missed_Apmt	Whether the patient missed the appointment.
Neighbourhood_Number	Where the appointment takes place.
Paper/Citation: Roberts, J. (2026) 'Gender Bias In Healthcare: Missed Appointment Prediction,' <i>CIS4057-N-FJ1-2025: Artificial Intelligence Ethics and Applications</i> . Teesside University. Unpublished. License: Not licensed. Questions: e5147249@live.tees.ac.uk	
INTENDED USE	
Primary intended uses: Predicting appointments that will be missed, in order to double-book or provide additional support. Primary intended users: Clinical settings in the following neighbourhoods of Brazil: ANDORINHAS, ANTÔNIO HONÓRIO, ARIOVALDO FAVALESSA, BARRO VERMELHO, BELA VISTA, BENTO FERREIRA, BOA VISTA, BONFIM, CARATOÍRA, CENTRO, COMDUSA, CONQUISTA, CONSOLAÇÃO, CRUZAMENTO, DA PENHA, DE LOURDES, DO CABRAL, DO MOSCOSO, DO QUADRO, ENSEADA DO SUÁ, ESTRELINHA, FONTE GRANDE, FORTE SÃO JOÃO, FRADINHOS, GOIABEIRAS, GRANDE VITÓRIA, GURIGICA, HORTO, ILHA DAS CAIEIRAS, ILHA DE SANTA MARIA, ILHA DO PRÍNCIPE, INHANGUETÁ, ITARARÉ, JABOUR, JARDIM CAMBURI, JARDIM DA PENHA, JESUS DE NAZARETH, JOANA D'ARC, JUCUTUQUARA, MARIA ORTIZ, MARUÍPE, MATA DA PRAIA, MONTE BELO, MÁRIO CYPRESTE, NAZARETH, NOVA PALESTINA, PARQUE MOSCOSO, PIEDADE, PRAIA DO CANTO, PRAIA DO SUÁ, REDENÇÃO, REPÚBLICA, RESISTÊNCIA, ROMÃO,	

SANTA CECÍLIA, SANTA CLARA, SANTA HELENA, SANTA LUÍZA, SANTA LÚCIA, SANTA MARTHA, SANTA TEREZA, SANTO ANDRÉ, SANTO ANTÔNIO, SANTOS DUMONT, SANTOS REIS, SEGURANÇA DO LAR, SOLON BORGES, SÃO BENEDITO, SÃO CRISTÓVÃO, SÃO JOSÉ, SÃO PEDRO, TABUAZEIRO, UNIVERSITÁRIO, VILA RUBIM.

Out-of-scope use cases:

Prioritising waiting lists. Clinical settings in any neighbourhoods not in the list above.

FACTORS

Data collection location: Brazil

Source: Real data from public health sector.

Number of individual patients: 62,146

M/F % split: 36:64

Age range: 0-102

Race: No data

Date range (bookings): 10-11-2015 to 08-06-2016 (~ 7 months)

Date range (appointments): 29-04-2016 to 08-06-2016 (~ 5 weeks)

METRICS

Model performance measures: Accuracy, Recall, Precision.

Fairness tests (gender): Equal Accuracy, Demographic Parity, Equal Opportunity

Decision thresholds: Protected/Privileged ≤ 0.8 for Equal Accuracy and Demographic Parity (Disparate Impact). Privileged – Protected (Equal Opportunity Difference) < 0.05 for Equal Opportunity.

EVALUATION DATA

Dataset: Hoppen, J. and Aquarela Analytics (2016) *Medical Appointment No Shows: Version 5*. Available from:

<https://www.kaggle.com/datasets/joniarroba/noshowappointments> (Accessed: 28 Jan 2026)

Size: Fourteen features and 110,527 instances.

Motivation: Real data from public sector medical appointments.

Split: 30% random (stratified) sample used for evaluation.

Three instances were removed due to missing Age (one -1 and two 115. According to https://en.wikipedia.org/wiki/List_of_Brazilian_supercentenarians four recorded Brazilians have reached 115, but none had reached that age in 2016 when the dataset was created.) and 221 from neighbourhoods with insufficient data (see INTENDED USE).

Features were selected using Mutual Information and Chi Square scores.

The test set was scaled using the Robust Scaler fit on the training set.

TRAINING DATA

As EVALUATION DATA (remaining 70%).

Initial distribution was 20% missed appointments. Training set balanced using oversampling.

QUANTITATIVE ANALYSES

	<i>Overall</i>	<i>Men</i>	<i>Women</i>
<i>Accuracy</i>	0.66	0.67	0.66
<i>Recall</i>	0.31	0.32	0.31
<i>Precision</i>	0.56	0.55	0.57

ETHICAL CONSIDERATIONS

Protected characteristics: Gender (Sex), Age, Disability.

No harms detected.

No mitigation applied.

CAVEATS AND RECOMMENDATIONS

N/A